

# Unit II

## Embedded IoT Platform Design Methodology

# INTERNET OF THINGS

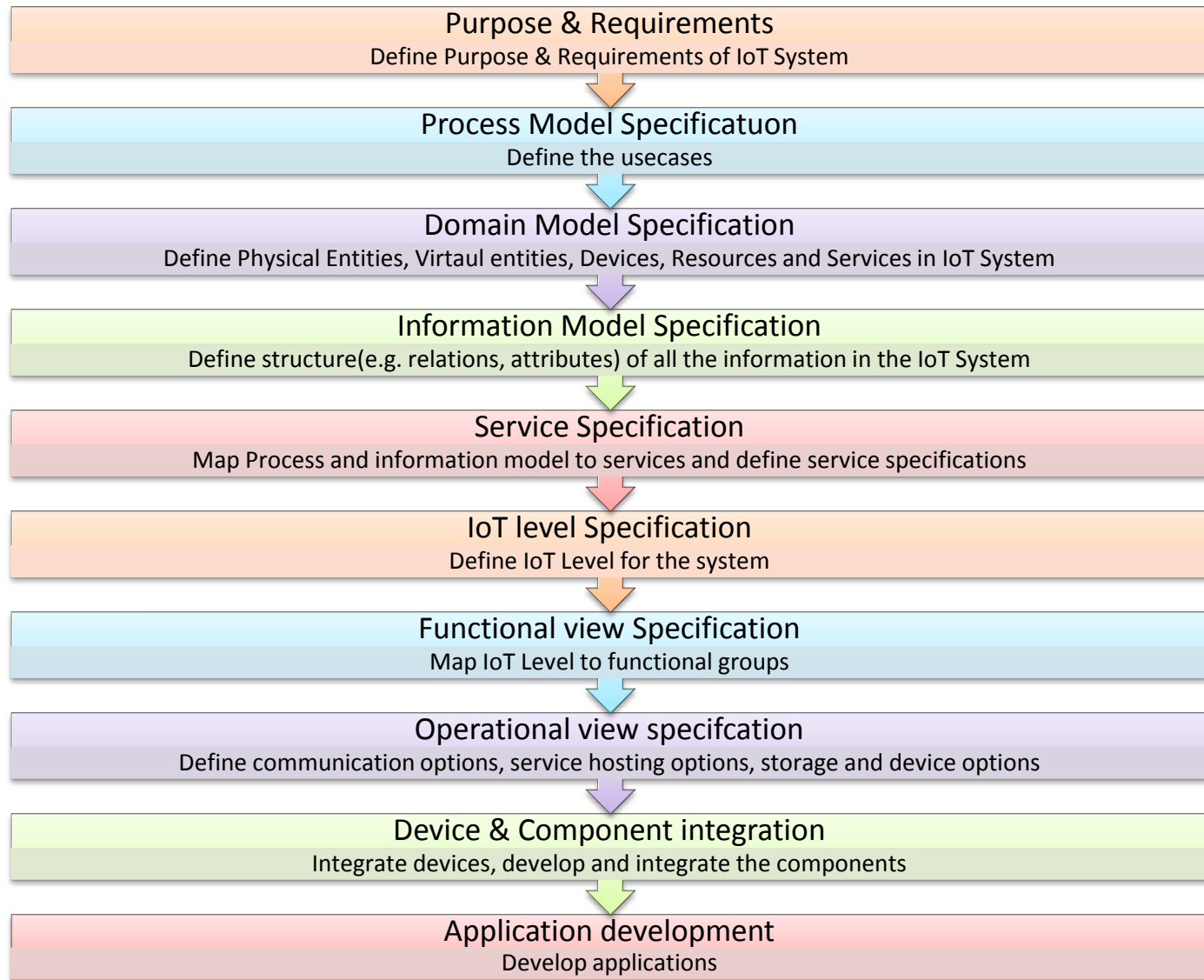
## A Hands-On Approach



# Outline

- Purpose and Requirement Specification
- Process Specification
- Domain model specification
- Information Model Specification
- Service specifications
- IoT Level Specifications
- Functional view specification
- Operational View Specification
- Device and Component integration
- Application development

# Steps involved in IoT System Design Methodology



# Advantages of Using Design methodology

- Reducing the design, testing and maintenance time
- Provide better interoperability
- Reduce the complexity

# Step 1 : Purpose and Requirement Specification

Defines

- **System purpose**
- **behavior and**
- **Requirements** (such as **data collection** requirements, **data analysis** requirement, **system management** requirements, data **privacy and security** requirements, **User interfaces** requirements)

# Step 1 : Purpose and Requirement Specification

- **Purpose** : An automated irrigation mechanism which turns the pumping motor ON and OFF on detecting the moisture content of the earth without the intervention of human
- **Behavior** : System should monitor the amount of soil moisture content in soil. In case the soil moisture of the soil deviates from the specified range, the watering system is turned ON/OFF. In case of dry soil, it will activate the irrigation system, pumping water for watering the plants.
- **System Management Requirements** : system should remotely provide monitoring and control functions

# Step 1 : Purpose and Requirement Specification

- **Data Analysis Requirements** : system should perform local analysis of data
- **Application Deployment Requirement** :  
Deployed locally on device, but acts remotely without manual intervention.
- **Security** : Authentication to Use the system must be available

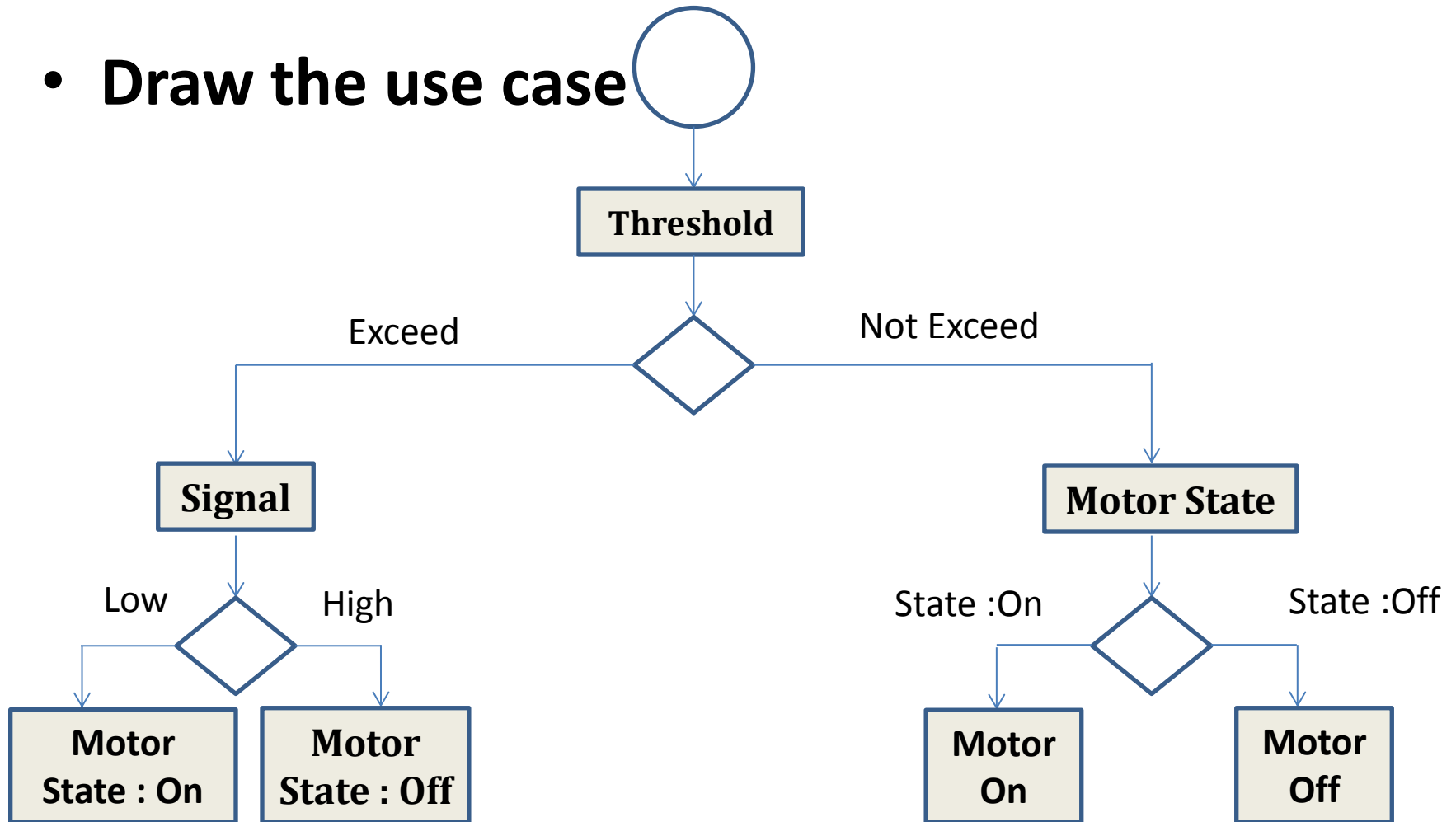
## Step 2 : Process Specification

- Define the process with the help of use cases
- The use cases are formally described based on Purpose & requirement specification
- In this use case :
  - **Circle** denotes a state or an attribute



# Step 2 : Process Specification

- Draw the use case



# Step 3 : Domain Model Specification

- **Describes the main concepts, entities and objects** in the domain of IoT system to be designed
- It defines the attributes of the objects and relationships between them
- Entities , Objects and Concepts include the following :  
Physical entity, Virtual entity , Device, Resource, Service

# Step 3 : Domain Model Specification

- **Physical Entity:**

- Discreet identifiable entity in physical environment
- For eg. Pump, motor, LCD
- The IoT System provides the information about the physical entity (using sensors) or performs actuation upon the Physical entity (like switching a motor on etc.)
- In smart irrigation example, there are three Physical entities involved :
  - Soil (whose moisture content is to be monitored)
  - Motor ( to be controlled)
  - Pump ( To be controlled)

- **Virtual Entity:**

- Representation of physical entity in digital world
- For each physical entity there is a virtual entity

# Step 3 : Domain Model Specification

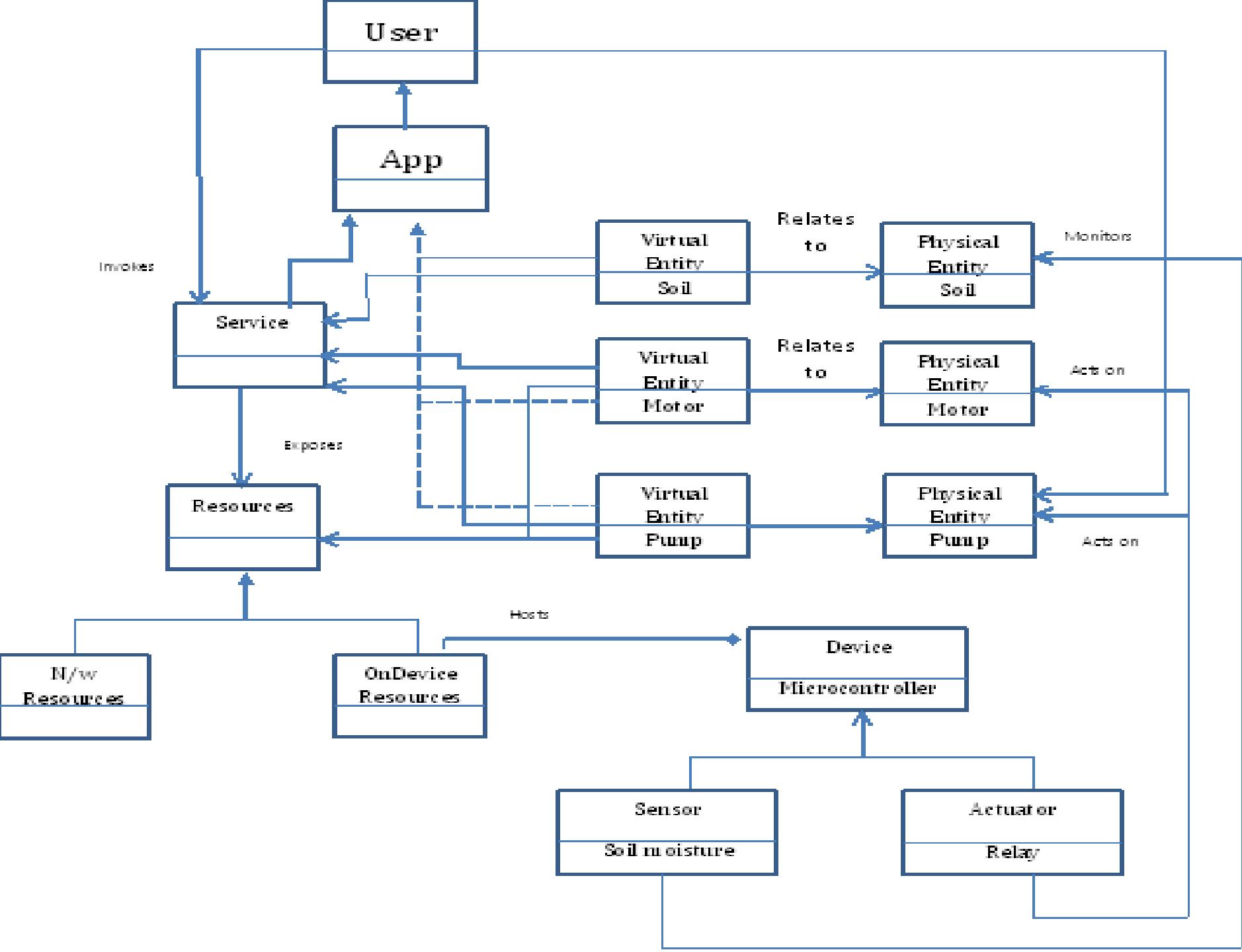
- **Device:**

- Medium for interactions between Physical and Virtual Entities.
- Devices (Sensors) are used to gather information from the physical entities
- Devices are used to identify Physical entities (Using Tags)
- In Smart Irrigation System, device is soil moisture sensor and buzzer as well as the actuator (relay switch) attached to it.

# Step 3 : Domain Model Specification

- In smart irrigation system there are three services :
  - A service that sets the signal to low/ high depending upon the threshold value
  - A service that sets the motor state on/off
  - A controller service that runs and monitors the threshold value of the moisture and switches the state of motor on/off depending upon it.

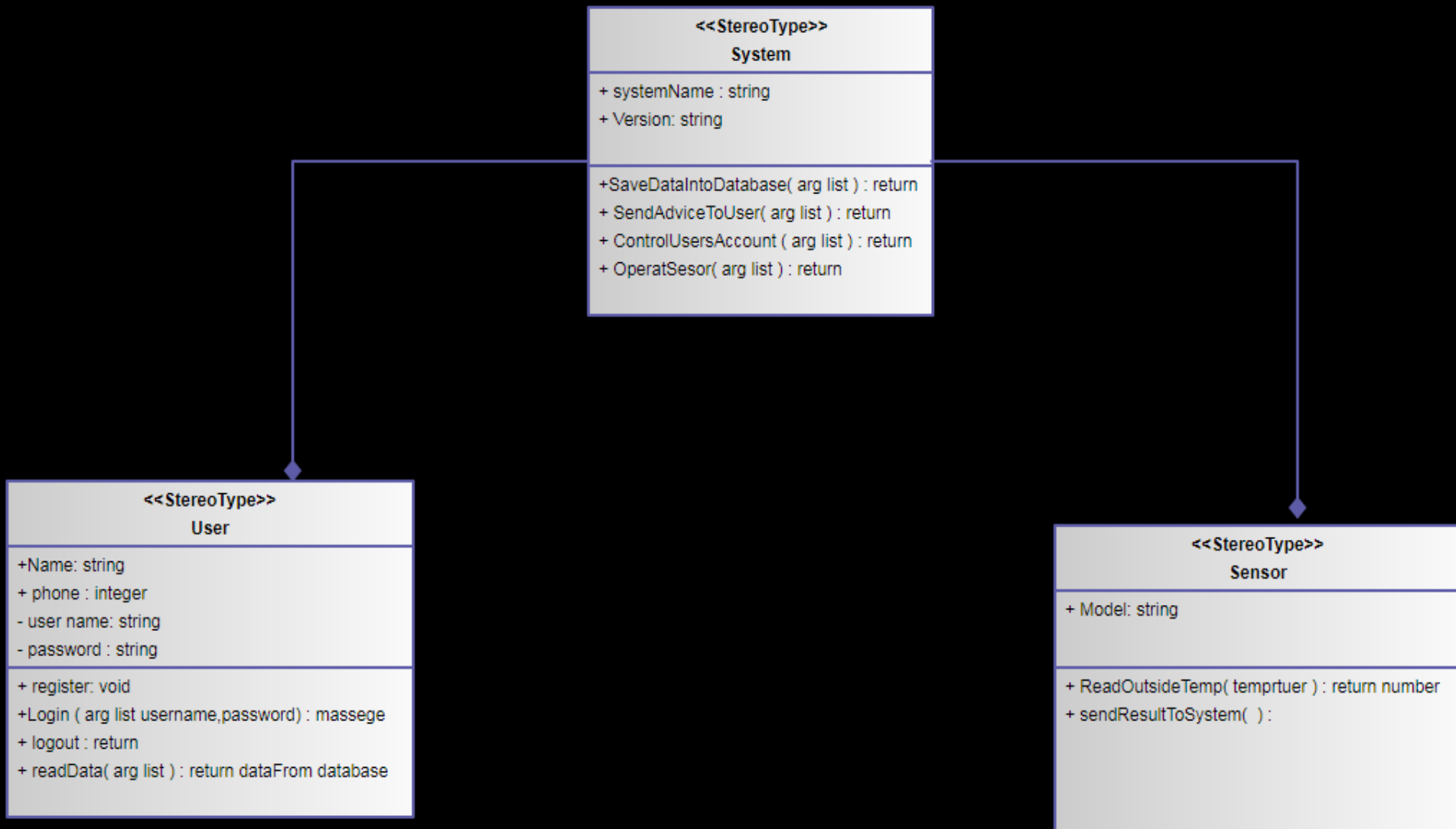
When threshold value is not crossed the controller retrieves the motor status from database and switches the motor on/off.



## Step 4 : Information Model Specification

- Defines the structure of all the information in the IoT system (such as attributes, relations etc.)
- It does not describe the specifics of how the information is represented or stored.
- This adds more information to the Virtual entities by defining their attributes and relations
- **I: e, Draw Class diagram**

# Step 4 : Information Model Specification





# Step 5 : Service Specification

- Define the services in IoT System, service **types**, service **inputs/outputs**, service **endpoints**, service **schedules**, service **preconditions** and service **effects**
- **Services can be controller service, Threshold service, state service for smart irrigation system**
- These services either change the state/attribute values or retrieve the current values.
- For eg.
  - Threshold service sets signal to high or low depending upon the soil moisture value.
  - State service sets the motor state : on or off
  - Controller service monitors the threshold value as well as the motor state and switches the motor on/off and updates the status in the database

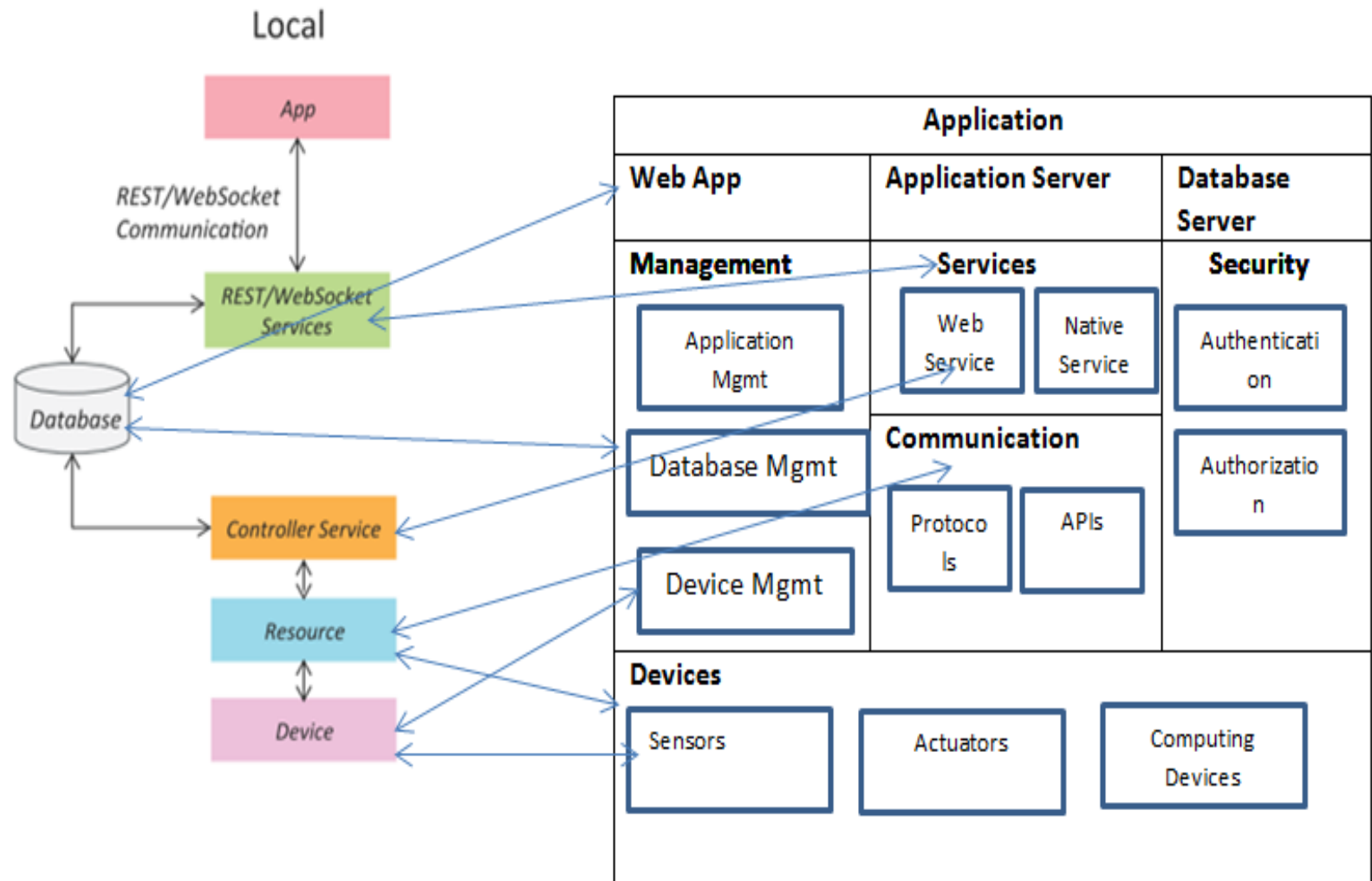
## Step 6 : IoT Level Specification

- Decide the deployment level of IoT System.  
Here I am using Deployment Level 1.

# Step 7 : Functional View Specification

- Define the functions of IoT System grouped into various functional groups.
- These functional groups provide functionalities for interacting with the concepts defined in Domain model specification.

# Step 7 : Functional View Specification



## Step 8 : Operational View Specification

- Define the Operations/options related to IoT System development
- Such as Device options, Storage options, Application hosting option

# Step 8 : Operational View Specification of automated irrigation system

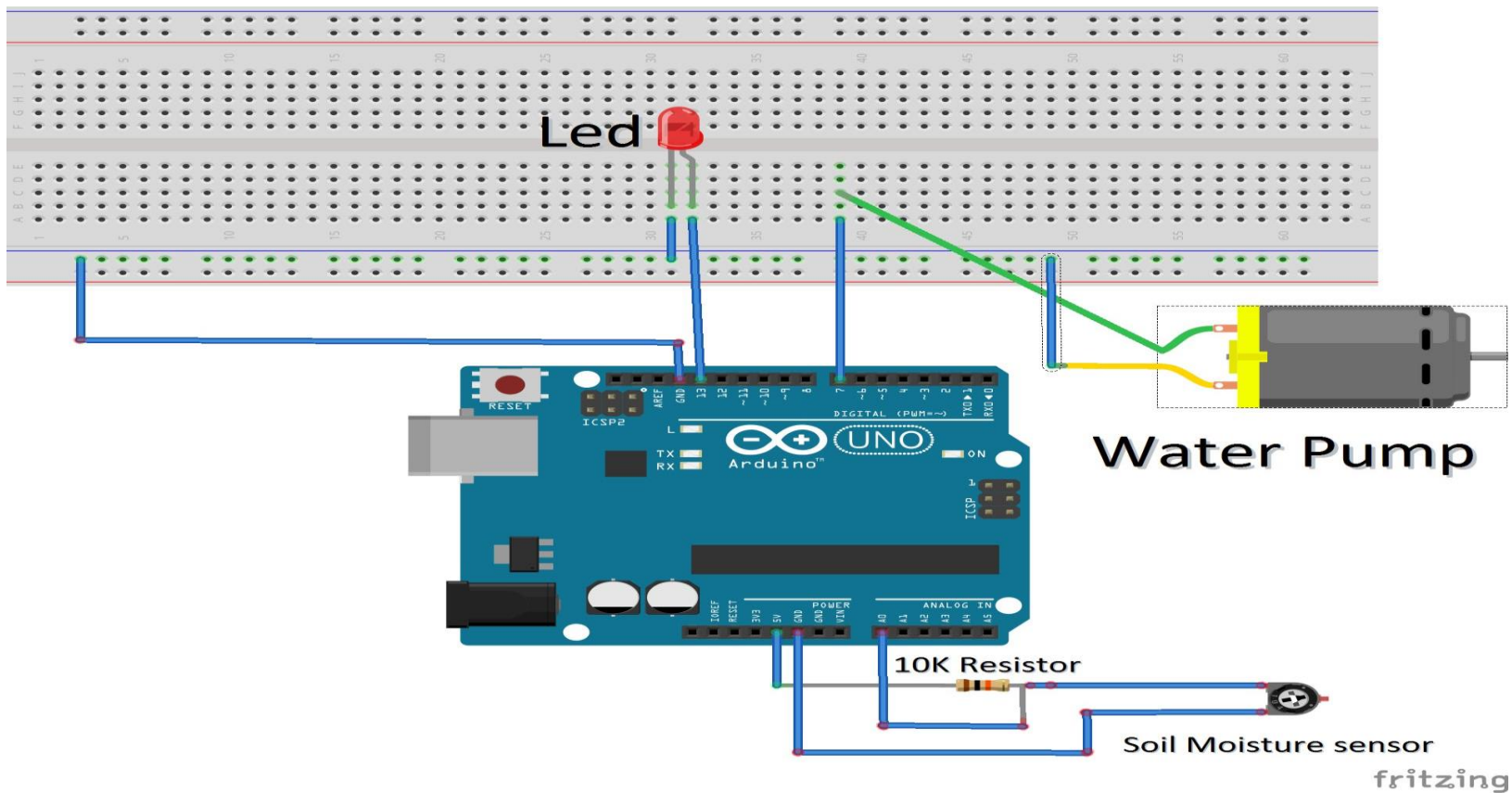
- Application
  - Web App : PhP WebApp
  - Application Server : Google App engine
  - Database Server : MySQL
- Services
  - Native : Controller Service
  - Web : REST
- Communication
  - Communication APIs : REST SPIs
  - Communication Protocol :
    - Link Layer: 802.11
    - N/w : IPV6
    - Transport : TCP
    - Application : HTTP

# Step 8 : Operational View Specification of automated irrigation system

- Management
  - Device Management: Arduino device management
  - Application Management : PHP App Management
  - Database Management: MySQL Db Mgmt
- Security
  - Login Management

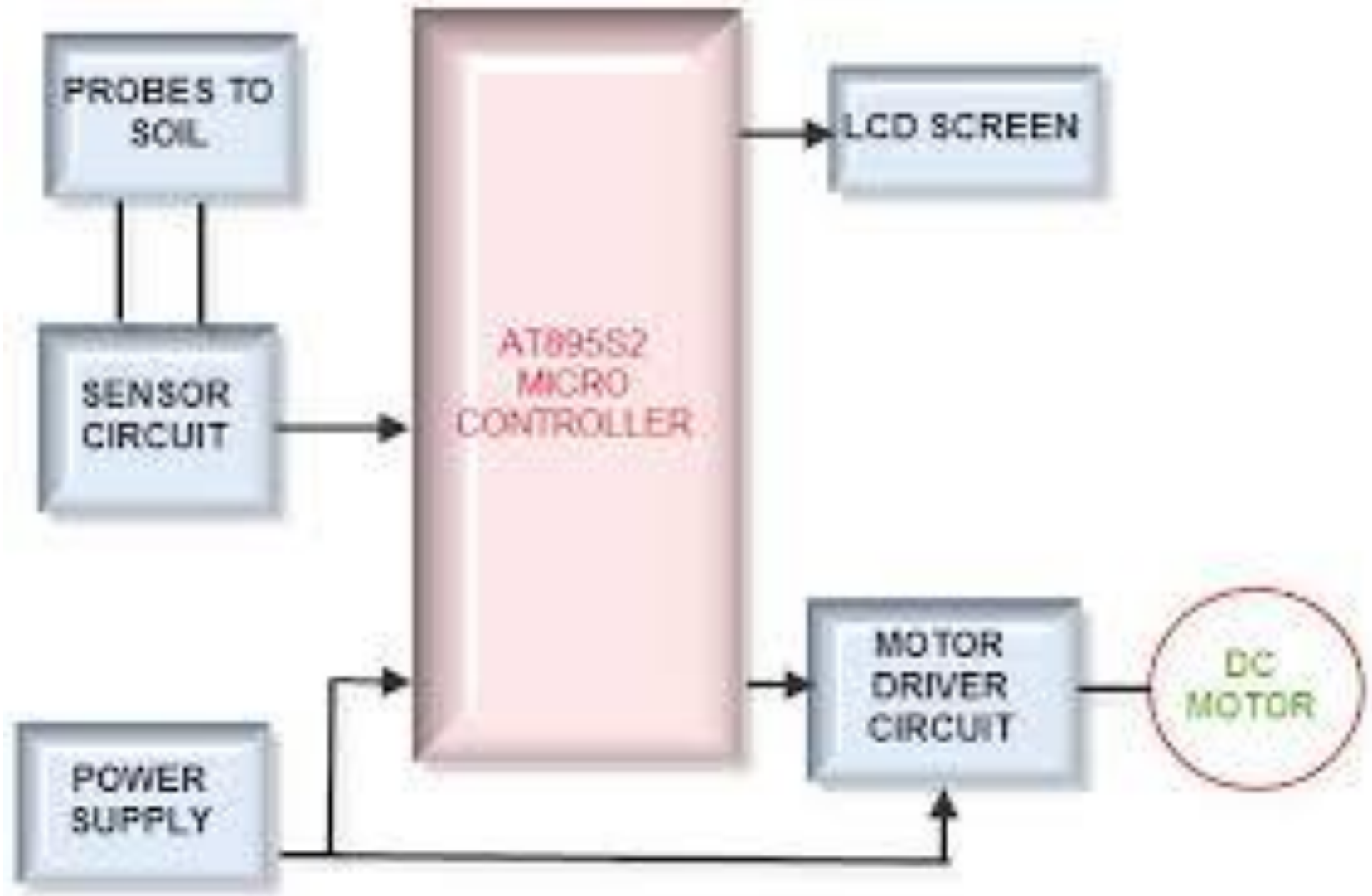
# Step 9 : Device and Component Integration

- Integrates the devices and components and draw a schematic diagram showing the same





# Device and Component Integration : Alternate Diagram



# Step 10 : Application development

- GUI / Screenshot of IoT Application